CCC-C-447b
June 6, 1963
SUPERSEDING
Int. Fed. Spec. CCC-C-00447a (Army-QMC)
February 16, 1962 and
Fed. Spec. CCC-C-447
October 14, 1960

FEDERAL SPECIFICATION

CLOTH, COTTON, PERCALE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cloth, cotton, percale.

1.2 Classification.

1.2.1 *Type.* The cloth covered by this specification shall be of one type.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

Federal Specifications:

CCC-T-191—Textile Test Methods.
PPP-P-51—Packaging, Packing, and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks and Synthetics).

Federal Standards:

Fed. Std. No. 4—Glossary of Fabric Imperfections.
Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 595—Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.)

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standard:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

Standard Color Card of America.

(Copies may be obtained from The Color Association of the United States, Inc., 200 Madison Avenue, New York 16, N. Y.)

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Rules and Regulations Under the Textile Fiber Products Identification Act.

(Copies may be obtained free of charge from the Federal Trade Commission, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is established, the dyed or bleached cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Yarn. The yarn shall be made from cotton which has been cleaned, carded, drawn and spun into single yarn (see 4.4) for both the warp and the filling.

3.3 Color. The color shall be bleached white or dyed as specified (see 6.2). The use of sulfur dyes or dyes containing elementary sulfur compounds capable of oxidation to sulfuric acid is prohibited. The dyestuffs shall be chosen and applied so that the dyed cloth shall show no more free or sulfide sulfur than the standard sample when tested as specified in 4.4. Unless otherwise specified, and no standard sample of the cloth is available, the following colors shall match the following numbers of Fed. Std. No. 595 and cable numbers of the Standard Color Card of America.

Fed. Std. No. 595		Standard Color Card of America	
Color	*Number*	*Color*	*Cable Number*
Green	24260	Blue	70023
Purple blue	35231	Brown	70107
Tan	30277	Light blue	70021
White	37875	Red	70080
Yellow	33695		

3.3.1 *Matching.* The dyed cloth shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.3.2 *Colorfastness.* The dyed and finished cloth shall show colorfastness to wet dry cleaning, crocking, laundering and light equal to or better than the standard sample. When no standard sample is established, the dyed and finished cloth shall show good colorfastness to wet dry cleaning, crocking, laundering, and light; if black or dark blue is specified, colorfastness to wet-crocking shall be not less than "fair". Testing shall be as specified in 4.4.

3.4 Physical requirements. The finished cloth shall conform to the requirements specified in table I when tested as specified in 4.4.

3.4.1 *Width.* The width shall be as specified (see 6.2) and shall be the minimum acceptable width exclusive of selvages.

3.5 Finish. The cloth shall be singed, bleached, or dyed and may be calendered. The residual size, finishing, and nonfibrous material content shall not exceed 2.5 percent when tested as specified in 4.4.

3.6 Shrinkage. The cloth shall be preshrunk and shall not shrink more than 1.0 percent in either the direction of the warp or the filling when tested as specified in 4.4. The preshrinking process used shall not be identified on the cloth, the ticket, or on the package.

3.7 Length. Unless otherwise specified (see 6.2), the cloth shall be in one continuous piece of not less than 40 yards.

3.8 Fiber identification. Each piece shall be labeled, ticketed, or invoiced in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.9 Workmanship. The cloth shall conform to the quality and grade of product established by this specification and the occurrence of defects shall not exceed the point level specified.

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TABLE I.—*Physical requirements*

Weight per square yard (minimum)	Yarns per inch (minimum)		Breaking strength (minimum)		Weave
	Warp	Filling	Warp	Filling	
Ounces			Pounds	Pounds	
3.0	85	72	45	30	Plain

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Examination of the end item. Examination of the end item shall be in accordance with the provisions of MIL-STD-105 and 4.2.1 through 4.2.4.

4.2.1 *Yard-by-yard examination.* The required yardage of each piece shall be examined on the face side and all defects as defined in Fed. Std. No. 4 which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.1.1, except that spots, stains and streaks shall be counted regardless of cause (i.e., not restricted to rust, oil or grease) and only those slubs and knots which exceed the maximum acceptable limits shown for each shall be scored (see 4.2.1.2). Defects on the selvage shall not be scored unless the body of the cloth is affected. No yard shall be penalized more than four points. For purposes of this examination, a yard shall begin at the last scored defect. The sample unit for this examination shall be one linear yard. The sample size shall be 750 linear yards, regardless of lot size. The sample yardage shall be examined from 25 pieces taken from 25 containers. An approximately equal number of yards shall be examined in each piece of the sample. The lot shall be unacceptable if the points per hundred square yards exceeds the following values:

30.00 points for dyed cloth
35.00 points for white cloth

Point computation shall be as follows:

$$\frac{\text{Total points scored in sample size} \times 3600}{\text{Sample size (yards)} \times \text{contracted width of material (inches)}} = \text{Points per 100 square yards}$$

4.2.1.1 *Demerit points.* Demerit points shall be assigned as follows:

For defects up to 3 inches in any dimension —one point.
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension —two points.
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension —three points.
For defects exceeding 9 inches in any dimension —four points.

NOTE: The following defects, when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy, or wavy cloth.

Objectionable odor.

Width less than specified.

Poor dye penetration, mottled, streaky, or cloudy.

Excessive neppiness.

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4.2.1.2 *Limits of slubs and knots.* Only slubs and knots which exceed the limits shown on figure 1 shall be scored.

4.2.2 *Examination for length.*

4.2.2.1 *Individual pieces.* During the yard-by-yard examination, each piece in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards below the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more pieces in the sample are defective in respect to length.

4.2.2.2 *Total yardage in sample.* The lot shall be unacceptable if the total of the actual lengths of pieces in the sample is less than the total of the gross lengths marked on the tickets.

4.2.3 *Examination for shade.* During the yard-by-yard examination, each piece in the sample shall be examined for shade. Any piece in the sample off shade, shaded side to side, side to center or end to end shall be cause for rejection of the entire lot represented by the sample.

4.2.4 *Examination for identification of preshrinkage process and noncompliance with Textile Fiber Products Identification Act.* During the yard-by-yard examination, each piece in the sample shall be examined for these defects. The lot shall be unacceptable if two or more pieces in the sample contain identification of the preshrinkage process by name or trademark on the cloth or ticket, or are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.3 Examination of preparation for delivery requirements. An examination shall be made to determine that packaging, packing and marking requirements of Section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be L-4 and the AQL shall be 4.0 defects per 100 units of MIL-STD-105.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.
	Preshrinkage process identified by name or trademark on container.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as: incomplete closure of case liners, container flaps; loose strapping, inadequate stapling.
	Bulging or distortion of containers.
Weight	Gross weight exceeds requirements.

4.4 Testing of the end item. The methods of testing specified in CCC-T-191, wherever applicable and as listed in table II shall be followed. The physical and chemical values specified in Section 3 except where otherwise specified, apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 3-1/2 continuous yards, full width of the finished cloth. The lot size shall be expressed in units of 1 yard. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified. The sample size shall be as follows:

Lot size in yards	Sample size
800 or less	2
801 to 22,000	3
22,001 and over	5

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TABLE II.—*Tests*

Characteristic	Requirement reference	Test Method
Yarn ply[1]	3.2	Visual
Weave[1]	Table I	Visual
Weight, ounces per square yard	Table I	5041
Yarns per inch	Table I	5050
Breaking strength, pounds	Table I	5100
Sulfur content[1]	3.3	4.4.1
Colorfastness to:		
Light	3.3.2	5660[2]
Laundering	3.3.2	5610
Wet dry cleaning	3.3.2	5622
Crocking	3.3.2	5651
Nonfibrous material content	3.5	2611
Shrinkage in laundering	3.6	5550[3]

[1] One determination shall be made from each sample unit and the results reported as "pass" or "fail."

[2] In the event of a dispute resulting from the test with method 5660 or as a result of suspected anomalous behavior of certain dye types or formulations the contracting officer may authorize the exposure to natural light in accordance with method 5662.

[3] The three specimens shall be selected from the cloth (sample unit) as follows: One specimen from each side of the cloth to within 3 inches of selvage and one specimen from the center of the cloth. No two specimens shall contain the same filling yarns. Each elongation value shall be prefixed with a minus sign and both inclosed in parenthesis.

4.4.1 *Test for sulfur content.*

4.4.1.1 *Solution (acid stannous chloride).* Dissolve 100 grams of stannous chloride crystals in 100 ml. of hydrochloric acid (35 percent concentration) and add 50 ml. of distilled water.

4.4.1.2 *Procedure.* One to 1.5 grams of the finished cloth, cut into small pieces, shall be used for the test. Place the sample in a test tube of suitable size and cover with stannous chloride solution. Place a filter paper which has been moistened with one drop of 5.0 percent lead acetate solution (made by dissolving 5 grams of reagent grade lead acetate to 100 ml. of distilled water; add a drop of glacial acetic acid to clear up if cloudy) over the mouth of the tube (punch needle hole in paper). Heat the test tube slowly over a low flame until the solution is boiling. Solution shall not be boiled for more than 10 to 15 seconds as prolonged boiling will reduce the sensitivity of the test. A brown to black stain appearing on the paper indicates the presence of sulfur.

4.4.1.3 *Evaluation.* The stained filter paper shall then be compared with that obtained from the standard. If the stained area is equal to or lighter than the standard in the degree of staining, the sample shall be considered satisfactory. If the degree of staining by the sample is in excess of the standard, the sample shall be considered unsatisfactory.

5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 shall be referred to for definitions and applications of the various levels of packaging and packing for civil agency procurements.)

5.1 **Packaging.** Packaging shall be Level A, B, or C, as specified (see 6.2 and 6.2.1).

5.1.1 *Levels A, B, and C.* The cloth put up as specified shall be packaged in accordance with the applicable requirements for class 3 fabrics of PPP-P-51.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (see 6.2 and 6.2.1).

5.2.1 *Levels A, B, and C.* The cloth shall be packed in accordance with the applicable requirements of PPP-P-51.

5.3 **Marking.** In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-51.

6. NOTES

6.1 Intended use. The cloth, cotton percale covered by this specification is intended for use in the manufacture of clothing.

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6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Color of cloth required (3.3).
(*c*) Width of cloth required (3.4.1).
(*d*) When length of piece is other than specified in 3.7.
(*e*) Selection of applicable levels of packaging and packing (5.1 and 5.2).

6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices, and alternatives as indicated in PPP-P-51 may be exercised in the preparation of delivery of cloth.

6.3 Standard sample. For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 Color (civil agencies). The colors designated in 3.3 are established to meet civil agency requirements for the following Federal Stock Numbers, as indicated:

Color	Federal Stock Numbers (FSN)	
Light blue	Turquoise blue	(8305-205-3066)
Brown	Dark brown	(8305-205-3067)
Purple blue	Copenhagen blue	(8305-205-3068)
Blue	Peacock blue	(8305-205-3069)
Tan	Light tan	(8305-205-3070)
Green	Green	(8305-205-3071)
Red	Dark red	(8305-205-3167)
White	White	(8305-205-3168)
Yellow	Yellow	(8305-205-3169)

6.5 Transportation description. The transportation description and minimum weights applicable to this commodity are:

Rail:
Cotton piece goods, finished, not otherwise indexed by name, in the original piece.
Carload minimum weight — Any quantity.

Motor:
Cotton piece goods not otherwise indexed, in the original piece.
Motor volume minimum weight— Any quantity.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:

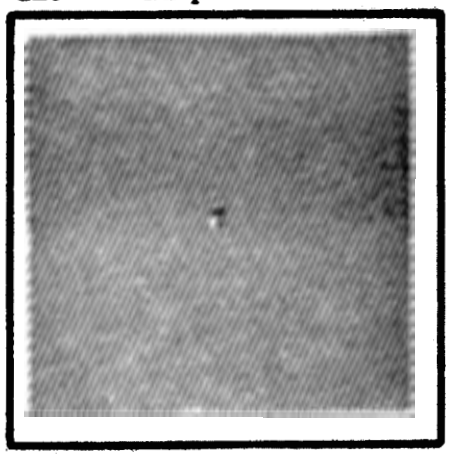
Army—GL

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Air Force—MAA

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Knot or Lump



Yarn Thickening (Slub)

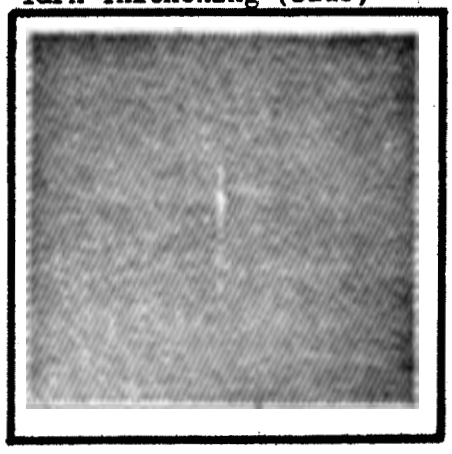


Figure 1.- Limits of acceptability.

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